

Post hoc bounds for heterogeneous tests

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Multiple testing setting

Formal setting of multiple testing - notations

Let $X \in \mathcal{X}$ be a sample of distribution $\mu \in \mathcal{P}$ where $\mathcal{P}$ is a set of probability measures. Define :

- $H_1, \dots, H_m \subset \mathcal{P}$, m null hypotheses,
- For all i , a p -value $p_i(X)$ such that, if $\mu \in H_i$, $F_i(\alpha) := \mathbb{P}_{X \sim \mu}(p_i \leq \alpha) \leq \alpha$ for all $\alpha \in [0, 1]$,

Goal : find a "good" rejection set $R(X)$ under some error constraints.

Notations :

- $\mathcal{H}_0(\mu) = \{i : \mu \in H_i\}$,
- $m_0 = |\mathcal{H}_0(\mu)|$,
- $V(R) = |\mathcal{H}_0(\mu) \cap R|$.

Binomial tests : Suppose $X = (X_1, \dots, X_m)$, where $(X_i)_{1 \leq i \leq m}$ are independent variables such that $X_i \sim \mathcal{B}(n_i, \pi_i)$ under H_i . Then :

- $p_i(X_i) := \mathbb{P}_{Y \sim \mathcal{B}(n_i, \pi_i)}(Y \leq X_i) = \sum_{j=0}^{X_i} \binom{n_i}{j} \pi_i^j (1 - \pi_i)^{n_i-j}$ is a p -value for i ,
- For all i , p_i is a discrete p -value and if the couples (n_i, π_i) are different, the respective cdf F_i are.

Other examples:

- Fisher's exact tests,
- Genome-wide association study.

Post hoc bounds

Definition 1 (Post hoc bounds)

Let $\alpha \in (0, 1)$. A post hoc bound of level α is a statistic $\hat{V}_\alpha : \mathcal{P}(\mathbb{N}_m) \rightarrow \mathbb{N}$ such that, for all $\mu \in \mathcal{P}$,

$$\mathbb{P}_{X \sim \mu}(\forall S \subset \mathbb{N}_m, V(S) \leq \hat{V}_\alpha(S)) \geq 1 - \alpha.$$

- Allows user to choose freely their rejection set.

Definition 2 (Joint Error Rate)

For a reference family $\mathfrak{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ and $\mu \in \mathcal{P}$, the joint error rate is :

$$JER(\mathfrak{R}, \mu) := \mathbb{P}_{X \sim \mu}(\exists k \in \mathbb{N}_K : V(R_k) > \zeta_k).$$

- Post hoc bounds ideas restricted at our reference regions,
- Allows to compute a valid post hoc bound with interpolation (Blanchard et al., 2020, Durand et al., 2020).

Until the end of the presentation, we are in a **top- k setting** :

$$R_k = \{\sigma(i), 1 \leq i \leq k\} \text{ (where } p_{\sigma(1)} \leq \dots \leq p_{\sigma(m)}).$$

Bretagnolle's inequality and top-k post hoc bounds

Proposition 1 (DKWM inequality - Improved version from Reeve, 2024)

Let $X_1, \dots, X_m \stackrel{iid}{\sim} F$ and $\hat{F}_m(t) := \frac{1}{m} \sum_{1 \leq i \leq m} \mathbb{1}_{X_i \leq t}$ for all $t \in \mathbb{S}$.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} (\hat{F}_m - F)(t) > \lambda\right) \leq \exp(-2\lambda^2).$$

When $(p_i)_{1 \leq i \leq m} \stackrel{iid}{\sim} \mathcal{U}([0, 1])$, $F(t) = t$.

Proposition 2 (Bretagnolle's inequality, Shorack and Wellner, 1986 with the improvement from Reeve, 2024)

Let $X_1, \dots, X_m$ be independent, denote by F_i the cdf of X_i and $\bar{F}$ the mean of F_i 's.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} (\hat{F}_m - \bar{F})(t) > \lambda\right) \leq e \cdot \exp(-2\lambda^2).$$

Theorem 1 (Bretagnolle's top- k calibration)

Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$ and

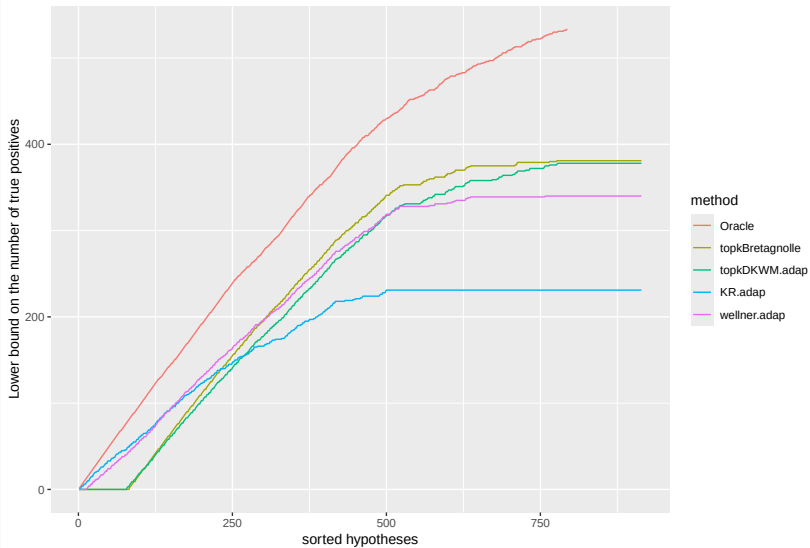
$$\zeta_k = k \wedge \left\lfloor m \bar{F}(p_{(k)}) + \sqrt{m} \lambda \right\rfloor.$$

Then, the reference family $\mathfrak{R} = ((R_k, \zeta_k))_{k \in \mathcal{K}}$ has its JER bounded by α .

Sketch of proof:

- $|R_k \cap \mathcal{H}_0| \leq \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq p_{(k)}},$
- Evaluate Bretagnolle's inequality at point $p_{(k)},$
- Bound $\sqrt{m_0}$ by $\sqrt{m}$ and $\sum_{i \in \mathcal{H}_0} F_i(p_{(k)})$ by $\sum_{1 \leq i \leq m} F_i(p_{(k)}).$

Simulations



Local test and estimation of m_0

Let $\alpha \in [0, 1]$ and, for $A \subset \{1, \dots, m\}$, $H_A := \bigcap_{i \in A} H_i$ the intersection hypothesis for A .

Suppose to have at hand, for all $A \subset \{1, \dots, m\}$, a local test φ_A (a test for H_A), and denote $\mathcal{N}(X) = \{A \subset \mathbb{N}_m : \varphi_A(X) = 0\}$.
 $\mathcal{N}(X)$ is the set of potential candidate for $\mathcal{H}_0(\mu)$.

Proposition 3 (Inversion procedure, Genovese and Wasserman, 2006)

Suppose that $X \sim \mu$ and $\varphi_{\mathcal{H}_0(\mu)}$ is a test of level α . Then, the statistic $\hat{V}_{\mathcal{N}}$ defined by $\hat{V}_{\mathcal{N}}(S) := \max_{A \in \mathcal{N}} |A \cap S|$ is a post hoc bound of level α .

- **Advantage** : Best bound on $V(S)$ from local tests,
- **Drawback** : Exponential computational cost.

Inversion procedure and concentration inequalities

Suppose that for all $A \subset \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

Example 1

For Bretagnolle's inequality,

$$\varphi_{A,t} = \mathbb{1} \left\{ \sqrt{|A|} (\hat{F}_A - \bar{F}_A)(t) > \lambda \right\} = \mathbb{1} \left\{ \sum_{i \in A} \mathbb{G}_i(t) > \sqrt{|A|} \lambda \right\},$$

where $\mathbb{G}(t) = (\mathbb{1}_{\{p_i \leq t\}} - F_i(t))_{1 \leq i \leq m}$.

Lemma 1

For all $S \subset \mathbb{N}_m$,

$$\hat{V}_{\mathcal{N}}(S) = \max \left\{ \ell \in \{0, \dots, |S|\} : \min_{A: |A \cap S| = \ell} \sup_{t \in [0,1]} \varphi_{A,t} = 0 \right\}.$$

Shortcuts and application to Bretagnolle's inequality

Lemma 2

Denote by $\varphi_{S,\ell}^{SC1} := \max_{t \in [0,1]} \min_{A: |A \cap S| = \ell} \varphi_{A,t}$, then :

$$\begin{aligned} \hat{V}_{\mathcal{N}}(S) &\leq \max \left\{ \ell \in \{0, \dots, |S|\} : \varphi_{S,\ell}^{SC1} = 0 \right\} \\ &\leq \min_{t \in [0,1]} \left\{ \max \left\{ \ell \in \{0, \dots, |S|\} : \min_{A: |A \cap S| = \ell} \varphi_{A,t} = 0 \right\} \right\}. \end{aligned}$$

Example 2

With Bretagnolle's local test, for $t \in [0, 1]$,

$$\min_{A: |A \cap S| = \ell} \varphi_{A,t} = \min_{0 \leq j \leq m - |S|} \mathbb{1} \left\{ \sum_{1 \leq i \leq \ell} (\mathbb{G}(t))_{(S:i)} + \sum_{1 \leq i \leq j} (\mathbb{G}(t))_{(S^c:i)} > \lambda \sqrt{\ell + j} \right\}.$$

- Reduces computation cost to $\mathcal{O}(m^2 \log(m))$.

Upper bound on m_0 via Bretagnolle's inequality

Theorem 2 (Bretagnolle's upper bound on m_0)

Let $\alpha \in (0, 1]$. Suppose that for all $k \in \mathcal{K}$, $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda = \sqrt{\frac{1 + \log\left(\frac{1}{\alpha}\right)}{2}}$, and

$$b(t) = \max \left\{ \ell : \ell^{-1/2} \left(\sum_{1 \leq j \leq \ell} (\mathbb{G}(t))_{(j)} \right) \leq \lambda \right\}, \text{ for all } t \in [0, 1].$$

Then, within the event $\{\mathcal{H}_0 \in \mathcal{N}\}$, we have $m_0 \leq \min_{t \in [0, 1]} b(t) := \hat{m}_0$.

Sketch of proof :

- Use Example 2 with $S = \{1, \dots, m\}$,
- Apply lemma 2, second inequality.

Adaptive calibration

Theorem 3 (Adaptive version of Bretagnolle's top- k calibration)

Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$, $\mathbb{F}(t) := (F_i(t))_{1 \leq i \leq m}$, $\hat{m}_0$ as the estimator of m_0 from Theorem 2 and

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \leq i \leq \hat{m}_0} \left(\mathbb{F}(p_{(k)}) \right)_{[i]} + \sqrt{\hat{m}_0} \lambda \right\rfloor^1.$$

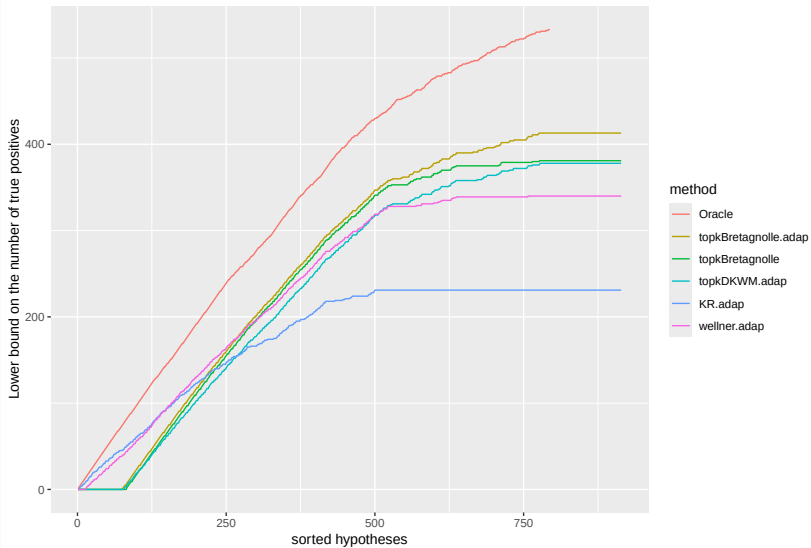
Then, the reference family $\mathfrak{R} = (R_k, \zeta_k)_{1 \leq k \leq m}$ has its JER bounded by α , where the R_k are the top- k regions.

Sketch of proof:

- Evaluate Bretagnolle's inequality at point $p_{(k)}$,
- Bound $\sqrt{m_0}$ by $\sqrt{\hat{m}_0}$ and $\sum_{i \in \mathcal{H}_0} F_i(p_{(k)})$ by $\sum_{1 \leq i \leq \hat{m}_0} \left(\mathbb{F}(p_{(k)}) \right)_{[i]}$ (worst case).

¹ $x_{[i]}$ is the i -iest larger coordinate from x .

Simulations



Summary :

- Applied a heterogeneous version of DKWM inequality in the context of post hoc MT,
- Get an inexact shortcut for the inversion procedure practically applicable to heterogeneous inequalities,
- Get an m_0 adaptive version of the resulting post hoc bound.

Perspectives :

- Heterogeneous k -FWER calibration (done),
- Get suitable heterogeneous versions of Wellner and K&R inequalities (work in progress).

Thanks a lot for your attention!

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Precisions on interpolated bounds

Proposition 4 (Interpolated bound, Blanchard et al., 2020)

Let $\mathfrak{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ a family with JER bounded by $\alpha \in [0, 1]$ and $\mathcal{A}(\mathcal{R}) := \{A \subset \mathbb{N}_m : \forall k \in \mathbb{N}_K, |R_k \cap A| \leq \zeta_k\}$.

Then $V_{\mathfrak{R}}^*(S) := \max_{A \in \mathcal{A}(\mathcal{R})} |R \cap A|$ is a level α post hoc bound.

Simes-like inequalities.

Simes-like inequalities

Proposition 5 (Simes inequality)

Suppose $(p_1, \dots, p_m)$ verifies the wPRDS property under $\mu \in \bigcap_{1 \leq i \leq m} H_i$. Then for all $\alpha \in [0, 1]$, $\mathbb{P}_{X \sim \mu}(\exists k \in \{1, \dots, m\} : p_{(k)} \leq \frac{\alpha k}{m}) \leq \alpha$.

Proposition 6 (Simes-like inequality derived from Döhler et al., 2018)

Suppose that, for $\mu \in \bigcap_{1 \leq i \leq m} H_i$, $p_1, \dots, p_m$ is a family of m independent p -values with known cdf $F_1, \dots, F_m$.

Let $\mathbf{H}(s, t) = \left(\frac{F_i(t)}{1 - F_i(s)} \right)_{1 \leq i \leq m}$ for all $(s, t) \in [0, 1] \times [0, 1]$, $\alpha \in]0, 1[$ and define

$\tau_m := \max \left\{ t \in \mathcal{A} : \sum_{1 \leq k \leq m} \mathbf{H}(t, t) \leq m\alpha \right\}$, and, for all $k \in \{1, \dots, m-1\}$,

$\tau_k := \max \left\{ t \in \mathcal{A}, t \leq \tau_m : (\mathbf{H}(\tau_m, t))_{[1]} + \dots + (\mathbf{H}(\tau_m, t))_{[m-k+1]} \leq k\alpha \right\}^2$.

Then, $\mathbb{P}_{X \sim \mu}(\exists k \in \{1, \dots, m\} : p_{(k)} \leq \tau_k) \leq \alpha$.

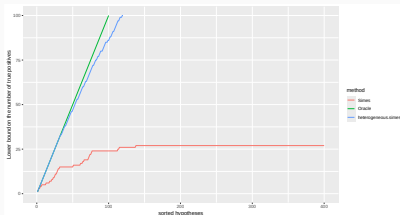
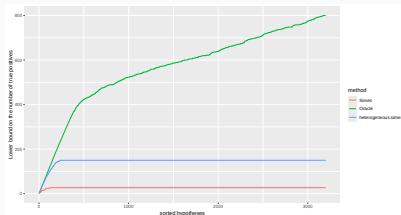
² $x_{[i]}$ is the i -iest larger coordinate from x .

Application to $k - FWER$ setting

k -FWER setting : $\zeta_k = k - 1$.

Theorem 4 (Simes-type calibration)

Let, for all $k \in \mathbb{N}_m$, $\zeta_k = k - 1$ and $R_k = \{i \in \mathbb{N}_m : p_i \leq \tau_k\}$. Then the reference family $((R_k, \zeta_k))$ has its JER bounded by α .



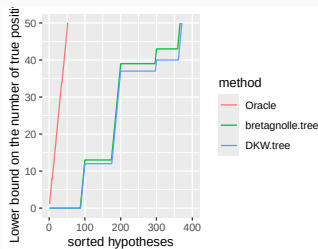
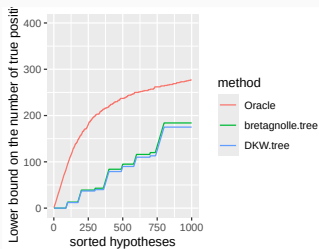
Simulations settings

- **top- k** : $m = 3200$ and $m_0 = 2400$. Continuous p -values sampled from a Gaussian model, with mean equal to 0 under H_0 and 3 under H_1 . Each p -value is discretized using a support of length 50 sampled from a uniform law.
- **k-FWER** : Same simulation setting with the support equal to 300.
- **Deterministic setting** : $m = 3200$ and $m_0 = 2800$, 8 active leaves of length 100 and each active leaf contains 50% of p -values under H_1 .

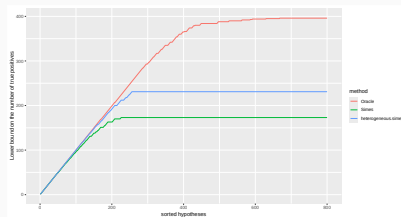
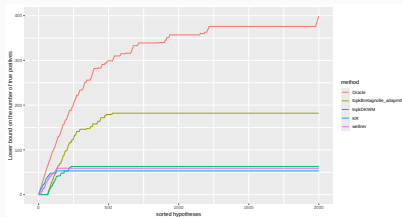
Same simulation settings than Döhler et al., 2018, with $N = 50$, $m = 2000$, $\pi_0 = 0.8$ and the response under H_1 simulated by binomials of parameter 0,25 (0,4 for Simes' simulations) against 0.1 under H_0 .

Deterministic setting simulations

Deterministic setting : the R_k are deterministic and have a forest structure, meaning that for all $(k, k') \in \{1, \dots, K\}^2$, $R_k \cap R_{k'} \in \{\emptyset, R_k, R_{k'}\}$.



Simulations with Fisher exact test



Complements on Uniform Wellner and K& R inequalities

Proposition 7

Let $\alpha \leq 0.31$. Then, in the top- k setting with p -values under the null independent,

$$\zeta_k := \left\lfloor \frac{\log(1/\alpha)}{\log(1 + \log(1/\alpha))} (1 + mp_{(k)}) \right\rfloor$$

gives a reference family with JER bounded by α .

Proposition 8 (Uniform version of Wellner inequality, Meah et al., 2024)

Let $U_1, \dots, U_m \stackrel{iid}{\sim} \mathcal{U}([0, 1])$ and $\kappa = \frac{\pi^2}{6}$. For all $\delta \in (0, 1]$,

$$\mathbb{P} \left(\forall t \in [0, 1], m^{-1} \sum_{1 \leq i \leq m} \mathbb{1}_{U_i \leq t} \leq th^{-1} \left(\frac{2 \log(\kappa/\delta) + 4 \log(1 + \log_2(1/t))}{mt} \right) \right) \geq 1 - \delta,$$

with $h(\lambda) = \lambda(\log \lambda - 1) + 1$.